Production Recommendation Pipeline Engineering Report

Student Roll No: `2025ag05921` **Dataset**: Kaggle E-Commerce Dataset (https://www.kaggle.com/datasets/abhayayare/e-commerce-dataset)
Workflow Framework: Modular Python DAG Orchestrator

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1. Executive Summary & Architecture Overview

This engineering report documents the implementation of an end-to-end, modular recommendation system processing clickstream events and explicit review metrics.

Our system incorporates a production-grade directory structure separating concerns across raw ingestion, strict schema validation, chronological partition splitting, multi-tier feature storage (SQLite + CSVs), machine learning, and automatic validation logging.

Pipeline Topology Flow:

1. **Ingestion**: Automated schema generator / Kaggle downloader writing to `data/raw/`.
2. **Validation**: Pandera engine isolating structural/logical violations to a `data/errors/` quarantine zone.
3. **Preparation**: Chronological train-test splitting (80/20 ratio) ensuring zero target or temporal leakage.
4. **Transformation**: Building sparse user-item matrix representations.
5. **Feature Store**: Calculating low-latency features stored within a SQLite RDBMS relational framework.
6. **Model Modeling**: Parallel execution of Bayesian Popularity Baseline and SVD Collaborative Filtering.
7. **Evaluation**: Performance matrix benchmarking reporting metrics directly to markdown formats.

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2. Data Validation Strategy

Data quality is protected by explicit validation schema rules built using the Pandera library. We check for: - **Null Constraint Violations**: Validates that critical entity identifiers (user_id, product_id) are present. - **Data Range Constraints**: Ensures review explicit rating feedback falls strictly within `[1, 5]` interval thresholds. - **Referential Integrity Constraints**: Verifies that any reviews submitted correspond to registered users in our database and matching inventory catalogs. - **Formatting Integrity Constraints**: Evaluates ISO formats across time-based interaction columns.

All failures are programmatically isolated into a quarantine CSV (`data/errors/quarantined_reviews.csv`) to prevent downstream bias and ensure maximum pipeline availability.

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3. Modeling Methodology & Findings

We trained and benchmarked two models representing different families of algorithms: 1. **Bayesian Popularity Recommender (Baseline)**: A simple non-personalized algorithm utilizing a prior-weighted popularity calculation to rank globally preferred catalog items. 2. **SVD Matrix Factorization (Personalized)**: A highly-personalized latent-factor model mapping users and items into a joint 50-dimensional space trained over 25 epochs.

Model Results Summary Table:

- **Popularity Recommender**: Precision@10: 0.0327 | Recall@10: 0.0241 | NDCG@10: 0.9102
- **Collaborative Filtering SVD**: Precision@10: 0.1179 | Recall@10: 0.1163 | NDCG@10: 0.9083

Key Interpretations:

- **Precision/Recall**: SVD outperforms the Popularity baseline by approximately **3.6x** (11.79% vs 3.27%), highlighting the value of capturing individualized user interest profiles.
- **NDCG@10 Consistency**: Both models achieve exceptional ranking relevance (~0.91), indicating highly optimal ordering among predicted recommendations.

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4. Operational Trade-offs & Limitations

While highly robust, the current architecture faces specific operational boundaries: - **SVD Cold-Start Problem**: If a new user signs up, the SVD model lacks collaborative history. To circumvent this, we implement a hybrid fallback route routing users directly to the Bayesian Popularity model scores. - **Storage Scaling**: Feature retrieval currently relies on a single SQLite instance. For hyper-scale production platforms, this should migrate to a dedicated Redis or Snowflake Feature Store cluster. - **Dynamic Retraining**: The DAG runs as a batch process. Real-time recommendation streaming pipelines would require Kafka or Spark Streaming additions.